

Efficiency of the Giant Panda and Snow Leopard Umbrella

Nicholas Chen

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Dr. Andrea DiGiorgio

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Abstract

This study evaluates the efficiency of the giant panda and the snow leopard as umbrella species to evaluate their viability as a flagship species in biodiversity conservation efforts in China. While previous studies have highlighted the public appeal and ecological value of a dual flagship species strategy, few have examined the spatial and economic efficiency of such an approach. This paper compares the number of species protected per square kilometer and the estimated cost of habitat maintenance per square kilometer for both species with the giant panda mainly serving as comparison. Using IUCN Red List data, the giant panda was found to protect approximately 0.071 species/km² at a cost of 9,884 USD/km², while the snow leopard protects about 0.002 species/km² at a cost of 100 USD/km². These results suggest that the giant panda is more efficient in terms of biodiversity protection per area, while the snow leopard offers a more cost-effective conservation investment. Despite the snow leopard's lower species coverage rate, its range includes more critically endangered and vulnerable species, as well as significant potential for ecosystem services. The findings indicate that while the giant panda remains an efficient conservation focus, incorporating the snow leopard may enhance the overall quality and reach of conservation outcomes.

Introduction

The giant panda is one of the most iconic species in biodiversity conservation and is considered invaluable in terms of providing environmental benefits. The giant panda is a “flagship species” which is a species responsible for raising public awareness and support for conservation. Flagship species are often large charismatic animals (Qian et al., 2020) or in lay terms, an animal with widespread appeal. As a flagship species, the giant panda has attracted

visitors from all over the world resulting in an increase in funding towards its conservation (Wei et al., 2018). The increased funding the giant panda generates as a flagship species should ultimately benefit numerous other species given that they also serve as an “umbrella species” (Simberloff, 1998). An umbrella species refers to a species whose conservation indirectly supports the protection of numerous other organisms within their ecosystem as their habitat range is large enough to overlap with numerous other species (Simberloff, 1998). For example, protecting giant pandas protects their habitat range, which in turn protects various other species whose habitat range overlaps with the giant pandas. More specifically, the giant panda’s habitat range “overlaps with 70% of the forest bird species, 70% of forest mammals, and 31% of forest amphibians that are endemic to mainland China” and “ninety-six percent of its range falls in the endemic centers (Fig. 2)” (B. V. Li & Pimm, 2016). Thus, they are considered an umbrella species providing vital support for the biodiversity of the endemic centers. In this study, the term umbrella species refers to the species’ ability to provide habitat protection for others, while flagship species refers to their symbolic and fundraising potential.

While the giant panda flagship species program may have worked for conservation programs so far, demonstrated in 2016 when the giant panda population in the wild was over 1800 moving them down to vulnerable status instead of endangered status (MODE, 2024), it may not work in the long run. For example, in protected areas of the giant panda range, there was an 81% loss of leopards, 38% loss of snow leopards, 77% loss of wolves, and 95% loss of dholes since the 1960s (S. Li et al., 2020) . These population declines bring into question the effectiveness of a single, giant panda, flagship species conservation program revealing a need to redefine old approaches or discover new approaches to biodiversity conservation. One such approach could be a dual flagship species program where another animal besides the giant panda,

could be regarded as a flagship species. In a study aimed at identifying potential flagship species in China's Three-River Source National Park, it was found that the snow leopard would be the best choice given that they performed the best in simulations involving Umbrella, Keystone, and Flagship indices, where Umbrella index measures habitat overlap, keystone measures predator-prey relationship, and flagship analyzes public interest in the species (Kau et al., 2025). Similarly, in a study on the potential impacts of making the snow leopard as a potential flagship species along with the giant panda, researchers found that the snow leopard not only has widespread public appeal, but also that the snow leopard's habitat range would better cover 284 different species' habitats than the giant panda (Yue et al., 2024) making them an ideal candidate as a second flagship species alongside the giant panda.

Building off of Yue et al. (2024) and Kau et al. (2025), this paper analyzes the efficiency of the giant panda and snow leopard—both of which qualify as an umbrella species and a flagship species—primarily as umbrella species, in terms of species coverage and cost per square kilometer, to evaluate their suitability as flagship species. Since the giant panda is already an established flagship species, their presence mainly serves as a comparison for the snow leopard. However, based on the research done by Yue et al. (2024) and Kau et al. (2025), it can be hypothesized that the snow leopard and giant panda will be equally efficient as umbrella species because while the snow leopard encompass a bigger range, they also cover more animals so the ratio of number of protected animals per square kilometers may be relatively similar. Ultimately, the act of making both the snow leopard and the giant panda a flagship species, would be a viable option for conservation.

Methods

To estimate the efficiency of these conservation campaigns, several variables and assumptions will need to be collected and made.

Coverage Per Square Kilometers

First, utilizing the IUCN Red List of Threatened Species, a list of threatened or endangered species whose habitat range overlaps with the giant panda habitat range was obtained. To do this, two tabs of the IUCN Red List were opened where one displayed the approximate range of the giant panda (Figure 1 left) and the other an empty map (Figure 1 right). On the empty map, using the IUCN polygon feature that comes with the map, a trace of the giant panda range was drawn to obtain a list of species living in the range of the giant panda. The same process was done with the snow leopard (Figure 2; left original, right drawn), but because the snow leopard range extends beyond China, so only the range that falls within China is considered. The types of species covered—particularly those that are vulnerable, endangered, or critically endangered in terms of conservation status—will also be discussed as efficiency in numerical terms does not account for the species that needs to be protected the most, a factor essential to conservation.

Errors and Assumptions

The drawings for each species habitat range include an error radius of ± 20 km (IUCN, n.d.) meaning that some species whose actual range does overlap with the giant panda and snow leopard, may not be accounted for and vice versa. This error can be negligible as many species' habitat ranges are relatively expansive, allowing some parts of the range to touch the drawn line and thus be accounted for in the calculation. Additionally, because the sample size is so large—greater than 1000—effects on the final coverage efficiency will be almost negligible. Moreover, this paper's analysis assumes species are evenly distributed across each species' entire

range. However, species density is rarely uniform due to terrain, climate, and human development factors, which may skew these calculations.

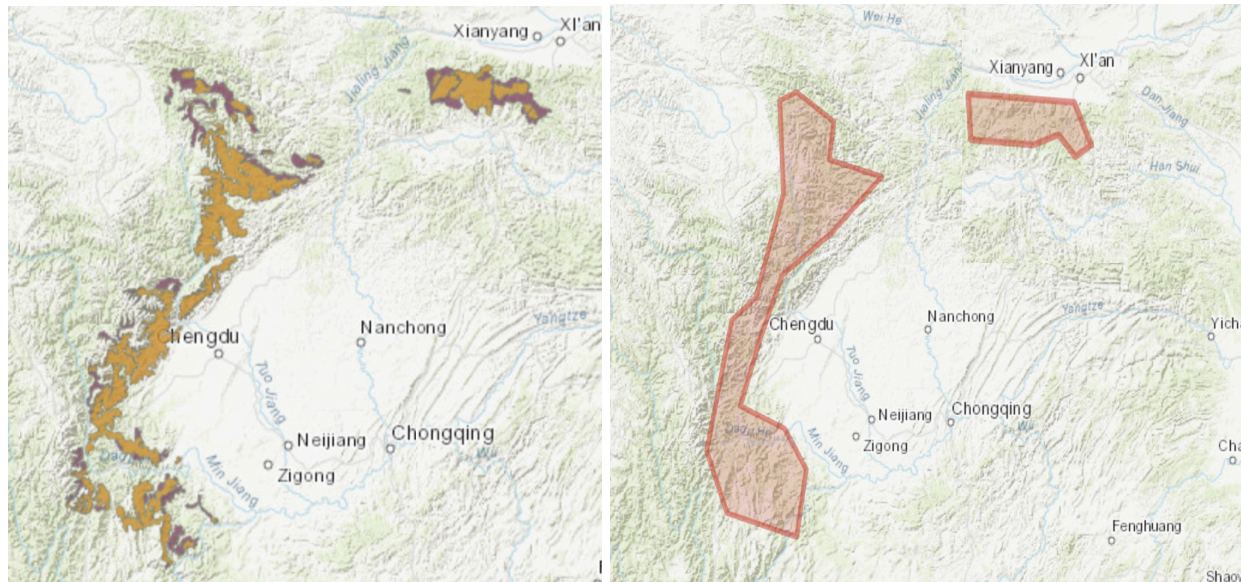


Figure 1. Screenshot from the IUCN Red List website showing the Habitat Range of the Giant panda (left). A trace of the giant panda habitat range (right). Source: IUCN Red List, <https://www.iucnredlist.org/species/712/121745669>, accessed April 9 2025.

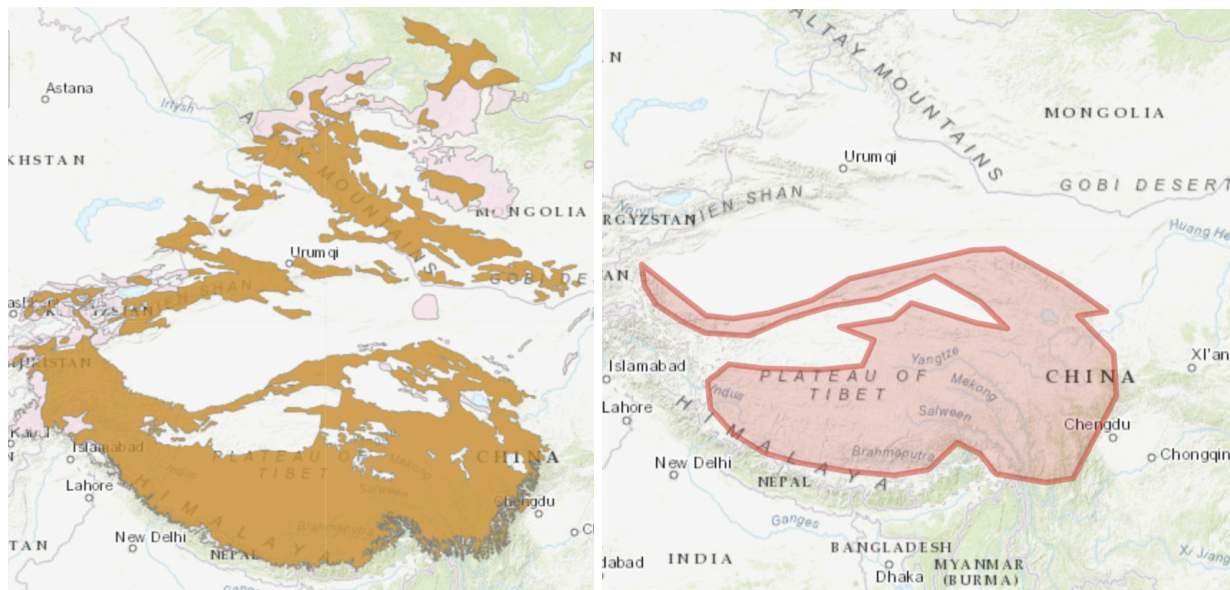


Figure 2. Screenshot from the IUCN Red List website showing the Habitat Range of the Snow Leopard (left). A trace of the giant panda habitat range using IUCN polygon feature (right).

Source: IUCN Red List, <https://www.iucnredlist.org/species/22732/50664030>, accessed April 9 2025

Estimation of Habitat area

Existing data was used to estimate the area each of these species occupies in China (km²) using Hu et al. (2024) for the giant panda and Snow Leopard Facts (n.d.) for the snow leopard.

Coverage efficiency for each species was determined using the formula:

$$\text{Species Covered Per km}^2 = \frac{\text{Number of Species Under Coverage}}{\text{Area of Coverage}}$$

Cost Per Square Kilometer

The exact cost needed to maintain the habitat for each species is very difficult to estimate because of uncertain development potential, opportunity costs, the need for comprehensive analysis of all project stages, and varying interests in the use of the land in question.

Additionally, some conservation organizations don't fully share their financial records. However, because investments in giant pandas have been occurring for many years, there have been numerous estimates on the cost of giant panda conservation. This paper will use Wei et al. (2018) estimated cost of giant panda conservation as it contains one of the most recent estimations on the cost of giant panda conservation. Wei et al. (2018) estimated that from 2011 to 2030, maintaining the giant panda habitat will cost approximately 255 million USD per year.

On the other hand, the conservation cost the Global Snow Leopard Ecosystem Recovery Program (GSLEP) projected that conservation of the snow leopard may cost from 150 to 200 million USD from 2013 to 2020 (*Saving Snow Leopards – Not Just a Day's Work*, n.d.). The

higher estimate of 200 million will be used so that it accounts for and allows room for unexpected costs, permitting better conservation preparation. However, this is the cost for the entire conservation program and the entire range of snow leopard, not limited to just habitat maintenance or just China, which necessitates modifying the estimate so that it applies only to China. It will be assumed that each square kilometer of land in the snow leopard range will be of equal value and almost all the projected cost will be used for habitat maintenance. This means that if 60% of the range falls into China, it would cost 60% of 200 million USD which is 120,000,000 USD. In actuality the cost would vary from a variety of factors including differing property values within each country's range and different resource requirements to maintain each section of the range, but because of limited time and data, it will be assumed that each area of land has equal value. To calculate the cost per square kilometer of land in each of the animal's habitat, the following formula will be used:

$$\text{Cost Per km}^2 = \frac{\text{Total Cost of Range}}{\text{Area of coverage}}$$

After the results are compiled, a plot will be made in RStudio Version 2024.12.1+563 (R Core Team, 2024; RStudio Team, 2024).

Results

The results came out to be that the giant panda habitat overlaps with approximately 1827 other different flora and fauna species and the snow leopard overlaps with approximately 2452 species. Both counts for each species do not include the species themselves. For the giant panda, it was estimated that their habitat covers approximately 25,800 square kilometers (Hu et al., 2024). For the snow leopard, it was estimated that they encompass "2,000,000 square kilometers, 60% of which is in China" (*Snow Leopard Facts*, n.d.). This means that the snow leopard occupies 1,200,000 square kilometers in China.

Using the previous data for each species and the formulas in the method section, the giant panda covers approximately 0.071 species per square kilometer and the snow leopard covers 0.002 species per square kilometer. Each square kilometer of land in the giant panda habitat range costs about 9884 USD while each square kilometer of land in the snow leopard range costs about 100 USD. Figure 3 gives a graphical representation of the contrast and Table 1 summarizes the types of species covered by the giant panda and the snow leopard.

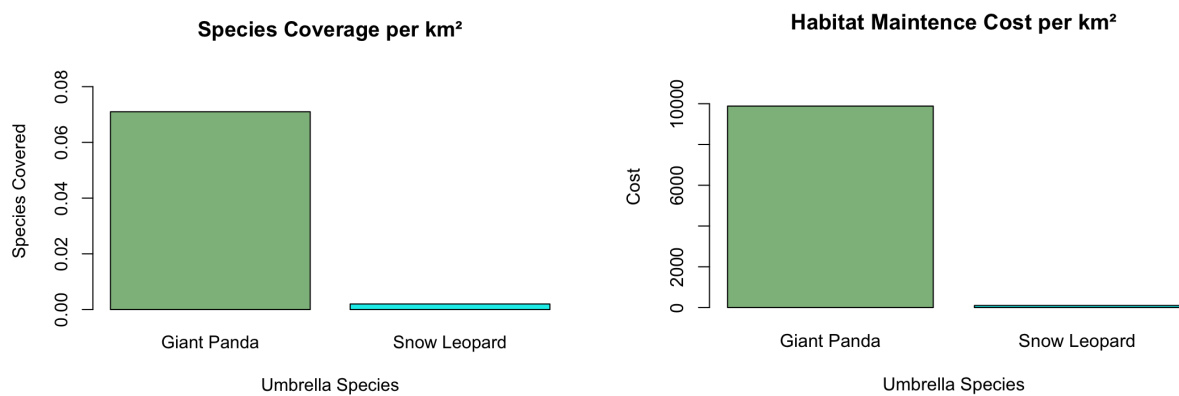


Figure 3 The giant panda covers 0.071 species per square kilometer, but costs 9884 USD to maintain the same amount of land, but the snow leopard covers 0.002 species per kilometer but costs about 100 USD to maintain the same amount of land. Chart made in Made in RStudio Version 2024.12.1+563 (R Core Team, 2024; RStudio Team, 2024).

Coverage by	Critically Endangered	Endangered	Vulnerable
Giant Panda	13	44	69
Snow Leopard	20	53	88
Both	9	28	53

Table 1 Table depicting the breakdown of critically endangered, endangered, and vulnerable species covered by the giant panda, snow leopard, and then both the giant panda and snow leopard. Data obtained from IUCN Red List <https://www.iucnredlist.org/search>

Discussion

The original hypothesis that both species would be equally efficient in being a flagship species is largely countered by the findings of the paper. From the results, it is clear that the giant panda, with its 0.071 species per square kilometer, is more efficient in protecting other species, than the snow leopard with its 0.002 species per square kilometer. The large difference in species per square kilometer may reflect habitat type rather than conservation inefficiency; giant pandas inhabit dense biodiversity hotspots (Li & Pimm, 2016), while snow leopards occupy harsher alpine regions (Yue et al., 2024) which may mean lower species richness. Therefore, the giant panda range, while encompassing less area, contains a significant amount of species resulting in the final species coverage value to be higher than the snow leopard. However, the snow leopard is more cost effective as it only requires 100 USD to maintain one square kilometer of land while the giant panda requires 9884 USD to maintain the same amount of land. In other words, the giant panda habitat costs more to maintain but protects more species, while the snow leopard costs less to maintain but protects less species.

Given the performance of the snow leopard as an umbrella species, it may not be a good idea to move forward with the dual flagship species program with the snow leopard as a second flagship species. Yue et al. (2024) suggested moving forward with the dual flagship species program because they primarily focused on the potential public and ecological advantages of using both species as flagships, emphasizing species overlap and appeal. Similarly, Kau et al. (2025) also focused on the snow leopard's potential public and ecological advantages given their

insight on the snow leopard as a flagship and umbrella species. However, this paper contributes new insight by quantifying economic and spatial efficiency, offering a complementary angle to their findings. Subsequently, the findings of this paper are crucial in the conversation on establishing a dual flagship species program with the snow leopard and the giant panda because it provides insight on the program from an efficiency and economic perspective, two aspects which are crucial for every country as resources are limited and therefore, proper management of resources is necessary.

Despite the results on efficiency analysis, conservation efforts must consider various other factors with one such factor being the conservation status of species protected by each animal's ranges. The giant panda habitat overlaps with approximately 13 critically endangered species, 44 endangered species, and 69 vulnerable species, (IUCN 2025). On the other hand, the snow leopard would protect 20 critically endangered species, 53 endangered species, and 88 vulnerable species (IUCN 2025). There are species overlap in these values as there are 9 critically endangered species, 28 endangered species, and 53 vulnerable species whose range overlaps with both the giant panda and the snow leopard. From this standpoint, it would make sense to make the snow leopard a flagship species because while it may not be as efficient in protecting other species, protecting their habitat range will offer higher quality protection, better protecting many critically endangered, endangered, and vulnerable species.

Despite the lack of efficiency of the snow leopards, it would be of great interest to establish them as a flagship species alongside the giant panda due to the low cost of maintenance and the better coverage of critical species. In 2010, giant panda conservation in Sichuan China generated an “ecosystem service” value of more than 20 times the investment (Wei et al., 2018, as cited in Yue et al. 2024) where ecosystem services refers to the economic worth of the

benefits, such as tourism, clean water, provisioning, and climate regulation, the ecosystem provides to humans. Likewise, the snow leopard may also generate an ecosystem service value of as high or even higher than the giant panda given the snow leopard's low cost of habitat maintenance. Given these considerations, it is reasonable to argue that conservation programs should transition to a different flagship species such as the snow leopard. Yet, the giant panda should still be kept as a flagship species as they generate valuable ecosystem services for China, is already a global icon for conservation with an example being the World Wildlife Fund's mascot as the panda, and billions of dollars have already been spent on the giant panda (Wei et al., 2018). Therefore, divesting in the giant panda and letting them go extinct will waste the efforts that have gone into protecting them. Thus, a dual flagship species program around the giant panda and snow leopard should be established.

Conclusion

This paper's findings highlight the difference in efficiency between the giant panda and the snow leopard as flagship species in terms of the number of species protected per square kilometer and the cost to maintain a square kilometer of land. The paper found that the giant panda, protecting 0.071 species per square kilometer, is more efficient in species protection than the snow leopard which protects 0.002 species per square kilometer. However, the giant panda costs more to maintain than a snow leopard with the former requiring 9884 USD for one square kilometer of land and the latter requiring only 100 USD for one square kilometer of land. While the snow leopard may be less efficient in species coverage, it costs less to maintain its habitats and if made a flagship species, its habitat range will better protect numerous high stake species. Therefore, it would benefit the environment and humans if the snow leopard becomes a flagship species given the potential ecosystem services they can provide.

This paper, while offering insight to the potential dual flagship species program through an efficiency and economic standpoint, has a few limitations. One of the biggest limitations is the lack of concrete costs when calculating the financial aspect of this paper. There are numerous differing reports on the cost of conservation for each species, making it difficult to choose the most appropriate one. Ultimately, the choice was made by considering the recency of the report and where the report came from. In future research, it would be desirable to develop a mathematical model on estimating the cost of conservation for different species. However, given market fluctuations and various factors such as land value, such approaches may be hindered. Additionally, the large difference between the species coverage of the giant panda and snow leopard may reflect habitat type rather than conservation inefficiency; giant pandas inhabit dense biodiversity hotspots (Wei et al., 2018), while snow leopards occupy harsher alpine regions (Yue et al., 2024) which may lower species count. So, while the giant panda habitat range is small, it contains a significant amount of species resulting in a higher species coverage value

Furthermore, this research paper builds off of Yue et al. (2024) and Kau et al. (2025) who already established that snow leopards are the ideal candidate as a second flagship species in Asia, but it would be beneficial to perform similar calculations on various other potential flagship species. The methods and calculations done in this paper can also be used worldwide to determine the efficiency of other flagship species throughout the world such as elephants, polar bears, and tigers. In this way, conservationists and organizations can determine if another flagship species is needed along with the already established species and provide insight into resource allocation.

This paper represents my own work in accordance with University regulations.
Nicholas Chen

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